

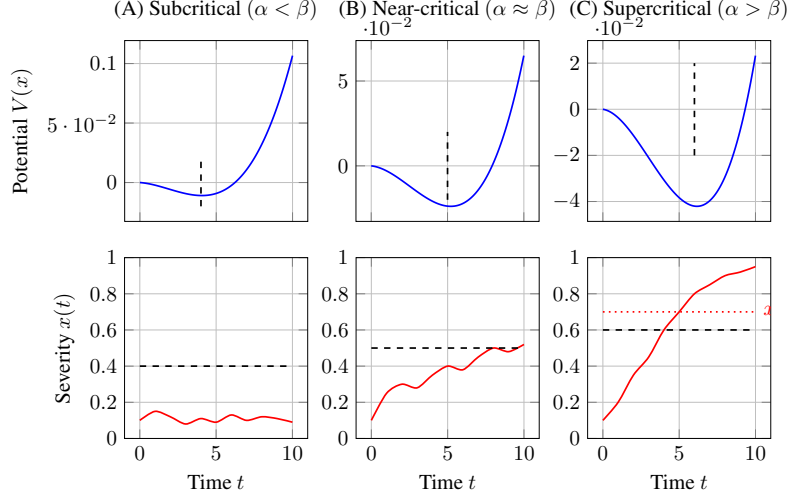


Figure 2: Self-amplifying bias dynamics in LLMs. Top row shows potential landscapes $V(x) = -\frac{\alpha}{2}x^2 + \frac{\alpha+\beta}{3}x^3 - \gamma x$ for different parameter regimes. Bottom row shows corresponding stochastic trajectories solving $dx(t) = \mu(x)dt + \sigma(x)dW(t)$, with drift $\mu(x) = \alpha x(1-x) - \beta x^2 + \gamma$ and noise $\sigma(x) = \sigma_0 + \sigma_1 x$. Dashed lines indicate critical thresholds x_c , and dotted line shows harmful threshold x_{harm} . Parameters: $\gamma = 0.01$, $\sigma_0 = 0.05$, $\sigma_1 = 0.1$.

4.2 Stationary Distribution $P_{\text{ss}}(x)$

When $\frac{\partial P}{\partial t} = 0$, (2) yields a *stationary distribution* $P_{\text{ss}}(x)$ satisfying

$$0 = -\frac{\partial}{\partial x} [\mu(x) P_{\text{ss}}(x)] + \frac{1}{2} \frac{\partial^2}{\partial x^2} [\sigma^2(x) P_{\text{ss}}(x)]. \quad (3)$$

Standard results give the closed-form expression [2]

$$P_{\text{ss}}(x) \propto \frac{1}{\sigma^2(x)} \exp\left(2 \int_0^x \frac{\mu(z)}{\sigma^2(z)} dz\right).$$

If μ strongly favors growth near $x > 0$, $P_{\text{ss}}(x)$ may concentrate away from zero, or even become *bimodal*. This is typically the sign of a *supercritical* regime, where high-severity states are stable attractors.

5 Critical Phenomena and Phase Transitions

5.1 Qualitative Picture of Criticality

Consider the drift parameters (α, β, γ) from the logistic-like example:

$$\mu(x) = \alpha x(1-x) - \beta x^2 + \gamma.$$

1. **If $\alpha < \beta$** , there is a stable fixed point near $x = 0$. Severity remains small, with the noise occasionally pushing it upward but the drift pulling it back. This regime is called *subcritical or aligned*.
2. **If $\alpha > \beta$** , the drift can remain positive after x surpasses some threshold, pushing it toward $x \approx 1$. This regime is called *supercritical or runaway*.

Near the boundary $\alpha = \beta$, the system may exhibit *critical slowing down* and increased fluctuations. In the Fokker–Planck landscape, P_{ss} can transition from unimodal (peaked at low x) to bimodal or peaked at high x .

5.2 Scaling Laws

We expect, from parallels with nonequilibrium phase transitions [4], that the correlation length ξ and relaxation time τ might diverge near criticality. Formally:

$$\xi(\Delta) \sim |\Delta|^{-\nu}, \quad \tau(\Delta) \sim |\Delta|^{-z\nu},$$

where $\Delta = \alpha - \beta$ measures how far the system is from the critical point. The exponents (ν, z) could, in principle, be measured by analyzing fluctuations of $x(t)$ in simulations or from real LLM logs.

6 First-Passage Analysis of Harmful States

A threshold $x_{\text{harm}} \in (0, 1)$ can be defined as the boundary beyond which the LLM’s outputs are deemed severely toxic or misaligned. The *first-passage time* T is:

$$T = \inf\{t \geq 0 : x(t) \geq x_{\text{harm}}\}.$$

One can derive a partial differential equation (PDE) for $\langle T \rangle(x)$, the expected time to blow-up starting from $x(0) = x$ [2]. In one dimension with SDE (1), the boundary condition is $\langle T \rangle(x_{\text{harm}}) = 0$, and one may impose $\frac{d\langle T \rangle}{dx}\big|_{x=0} = 0$ if $x = 0$ is reflecting. The solution generally shows an exponential sensitivity to integrals of μ/σ^2 , a hallmark of how quickly a supercritical drift can push $x(t)$ up to the harmful region.

7 Implications for Agents and Extended Reasoning

Beyond static text generation, modern LLMs can act as *agents*, performing multi-step reasoning or planning over extended time horizons. In such scenarios, the *severity* $x(t)$ can keep feeding back into the agent’s policy or chain-of-thought. If, for example, the system is in a supercritical domain of parameters, we risk *cascading bias* that leads to undesirable or disallowed outputs as the agent self-references prior negative statements.

Formal Verification Potential. A promising direction is using these SDE and Fokker–Planck equations for a *rigorous check* of whether, under all typical sampling dynamics, the severity distribution remains *stationary* near 0 or if it flows inevitably to high x . If we can bound $\mu(x)$ below x , or show that $P_{\text{ss}}(x)$ is unimodal at low severity, this might serve as a formal proof of *subcritical alignment* for an LLM-based agent. Conversely, detecting that x_{harm} is almost certainly reached within finite time would be a red flag, signifying *runaway misalignment* in extended inference.

Interpretation and Safety Gains. In practice, evaluating $\mu(x)$ and $\sigma(x)$ from real LLM data would require carefully controlled experiments and robust severity metrics. But should the fit reveal that the system is "near critical," design teams might reduce α (the self-amplification) or increase β (alignment damping) to ensure stable performance over many reasoning steps.

8 Conclusion and Outlook

This paper has presented a **stochastic differential equation framework** for modeling LLM chain-of-thought severity. By positing that severity $x(t)$ evolves in a near-Markov manner, the analysis shows how the *Fokker–Planck equation* naturally arises to describe the probability flow in severity space. Crucially, small changes in drift parameters can yield a *phase transition* from subcritical (safe or self-correcting) to supercritical (runaway) regimes. The work analyzes the stationary distribution, first-passage times to harmful thresholds, and near-critical scaling laws reminiscent of classical nonequilibrium physics.

Implications for extended reasoning models are profound: in principle, these equations open the door for *formal verification* of stability or guaranteed subcritical behavior. Coupled with improved severity metrics and data-fitting procedures, the approach could help LLM developers ensure that multi-step, agentic reasoning systems do not inadvertently *self-escalate* into severely misaligned outputs. Future work includes multi-dimensional expansions, memory kernels for more realistic references to older tokens, and bridging to interpretability methods that track which internal components of an LLM drive $\mu(x)$ at each stage.

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